Supplementary table 3: The completed GRADE checklist for overall quality assessment of evidence for metabolic risk factors review among adults in Ethiopia.

Outcomes	**Number of Studies**	**Risk of bias**	**Inconsistency**	**indirectness**	**imprecision**	**Publication bias**	**Certainty of evidence**
Hypertension	36	Not serious	Serious	Not serious	Not serious	Not serious	⊕⊕⊕⊕ Moderate
Overweight/ obesity	16	Not serious	Serious	Not serious	Not serious	Serious	⊕⊕⊕⊕ Low
Impaired glucose tolerance	14	Not serious	Serious	Not serious	Not serious	Serious	⊕⊕⊕⊕ Low
Metabolic syndrome	8	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and hypertension	17	Not serious	Serious	Not serious	Serious	Not serious	⊕⊕⊕⊕ Low
The association of sex and overweight/obesity	7	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and impaired glucose tolerance	3	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and metabolic syndrome	7	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and Hypertension	26	Not serious	Serious	Not serious	Not serious	Not serious	⊕⊕⊕⊕ Moderate
The association of age and overweight/obesity	9	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and impaired glucose tolerance	6	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and metabolic syndrome	6	Not serious	*	*	*	*	⊕⊕⊕⊕ Low

*Due to small number of included studies (n <10) could not be formally assessed.

Reasons to downrate certainty of evidence

1. **Risk of bias** was assessed by the JBIs critical appraisal tool for prevalence studies and categorized as follows:
   Not serious: if the included study in the analysis has low risk of bias score in the quality assessment
   Serious: judged between not serious and very serious or if some of the included study had high risk bias but lower to say very serious.
   Very serious: if more than 50% of the included studies had higher risk of bis score in the quality assessment.
2. **Inconsistency (heterogeneity)** of the included studies was based on the following criteria:
   Not serious: lower heterogeneity (<30%), Overlap of confidence intervals, close variance estimates across studies.
   Serious: if one of the following: high level heterogeneity (30-60 or above) or minimal overlap of confidence intervals or wider variance estimates across studies.
   Very serious: if two or all of the following: higher heterogeneity (30-60%) or above and minimal overlap of confidence intervals and/or wider variance estimates across studies.
3. **Indirectness** of the study was examined by assessing the concordance in population, exposure, and outcome (PEO's) of the study and the evidence
   Not serious: not have indirect comparison, not have different in terms of population, or use the same outcome measure
   Serious: if the studies have any of the following: indirect comparison, or different in terms of population, or different outcome measure
   Very serious: if the studies involved two or more of the following: indirect comparison, different in terms of population, and different outcome measure
4. **Precision** of the evidence was assessed by examining the width of the confidence intervals and associated effect of the exposure
   Not serious: if the confidence intervals of estimates are precise
   Serious: if the estimates involved wide confidence interval
   Very serious: if the estimates involved very wide confidence interval
5. **Publication bias** was assessed by observing the symmetrical distribution of studies in the funnel plot and conducting the Begg's and Egger's test of publication bias.
   Not serious: If study had symmetrical distribution in the final plot and tests did not show significant associations.
   Serious: If study had symmetrical distribution in the final plot but Begg's and Egger's tests showed significant publication bias.
   Very serious: if studies had not symmetrical distribution in the final plot, and also showed publication bias the Begg's and Egger's test.

Reasons to increase certainty of evidence

1. Large magnitude of effect
   None: if the effect estimate is <2
   Strong: if the effect estimate is >2 or <0.5
   Very strong: effect estimate >4 or <0.25
2. Effect potential residual confounding factors
   Dilute the effect: when the measurement is taken in some selected portion of participants and the actual measurement is expected to result higher estimates.
   Suggest strong effect: when residual confounding is expected to increase the effect but not measured.
3. Dose response gradient
   Yes: if there is dose gradient relationship
   No: if there is no dose gradient relationship